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by

Ritam Mondal

(23IM30018)

Under the supervision of

Dr. Abhishek Sharma



Department of Industrial and Systems Engineering

Indian Institute of Technology Kharagpur

Autumn Semester, 2026–27

November 2026

DECLARATION

I certify that

(a) The work contained in this report has been done by me under the guidance of my supervisor.

(b) The work has not been submitted to any other Institute for any degree or diploma.

(c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.

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DEPARTMENT OF INDUSTRIAL AND SYSTEMS ENGINEERING

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

KHARAGPUR – 721302, INDIA



CERTIFICATE

This is to certify that the project report entitled "**Risk-Aware Deep Reinforcement Learning for Daily Portfolio Allocation: Controlled Ablation of Temporal Memory, Reward Shaping, and Turnover Regularization**" submitted by **Ritam Mondal** (Roll No. 23IM30018) to Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Bachelor of Technology in Industrial and Systems Engineering is a record of bona fide work carried out under my supervision and guidance during Autumn Semester, 2026–27.

Date: November 10, 2026

Place: Kharagpur

Dr. Abhishek Sharma

Department of Industrial and Systems Engineering

Indian Institute of Technology Kharagpur

Kharagpur – 721302, India

Acknowledgement and Abstract

Acknowledgement:

I express my sincere gratitude to my supervisor, Dr. Abhishek Sharma, for his insightful guidance, intellectual mentorship, and continuous encouragement throughout this B.Tech. Project. His domain expertise in financial engineering, operations research, and sequential decision processes provided invaluable direction for this study. I also thank the Department of Industrial and Systems Engineering, Indian Institute of Technology Kharagpur, for providing computational facilities, and my peers and family for their continuous support.

Abstract:

Automating daily portfolio allocation via deep reinforcement learning (DRL) requires balancing return generation against market non-stationarity, partial state observability, and execution frictions. Prior literature separately suggests temporal encoders, risk-sensitive rewards, and turnover penalties, but these mechanisms are rarely evaluated under a single controlled setup. This report investigates a four-stage ablation: (i) Model M1: PPO with a feedforward MLP baseline and net return reward; (ii) Model M2: PPO with a recurrent LSTM encoder; (iii) Model M3: LSTM-PPO with recursive Differential Sharpe Ratio (DSR) reward; and (iv) Model M4: LSTM-PPO with DSR reward and a quadratic weight-change penalty ($\lambda_{\text{turn}} = 0.1$). All agents output continuous Dirichlet allocations over ten liquid National Stock Exchange of India (NSE) equities and an assumed zero-return cash reserve.

Models are evaluated across four chronological walk-forward folds spanning 2018–2025 across three random seeds under realistic proportional transaction costs (0.1%). In our experimental setup, the LSTM encoder (M2) improves upon the MLP baseline (M1) in pooled Sharpe ratio (0.92 vs. 0.71) and cumulative return (28.2% vs. 20.6%). The DSR reward (M3) demonstrates mixed, regime-dependent effects without an overall advantage over M2. Turnover regularisation (M4) effectively curtails daily turnover (cutting costs from 4.14% to 0.97% of capital) and achieves the shallowest maximum drawdown (-16.04%), but trades off cumulative return. Crucially, the passive Equal-Weight benchmark outperforms all RL models in pooled return (36.48%) and Sharpe ratio (1.13), driven by continuous broad-market exposure in a secular bull market and cash drag from the agents' learned liquidity holdings.

Keywords: Deep Reinforcement Learning, Portfolio Allocation, Proximal Policy Optimisation, Dirichlet Distribution, Differential Sharpe Ratio, Turnover Regularisation, Walk-Forward Validation.

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Chapter 1 — Introduction

1.1 Background and Motivation

Portfolio allocation is a classic problem in quantitative finance and operations research, concerning the sequential distribution of capital across multiple financial assets to achieve an attractive balance between risk and return. In real equity markets, asset prices exhibit low signal-to-noise ratios, non-stationary return distributions, and time-varying volatility clustering. Furthermore, an investor observing public price and volume records never perceives the complete economic state of the market, but only a noisy, lossy projection of it.

Sequential portfolio rebalancing is inherently path-dependent: the capital available at any decision step depends directly on past allocation choices, the path of intervening asset returns, and the cumulative friction of transaction costs. This sequential structure makes Deep Reinforcement Learning (DRL) an appealing framework, as DRL agents learn policies aimed at optimizing long-term cumulative returns under environmental uncertainty rather than relying solely on isolated, one-step point forecasts.

However, practical algorithmic execution requires careful accounting for trading frictions. Without appropriate regularization, DRL agents operating in continuous action spaces can reallocate excessively from day to day, incurring transaction fees that rapidly erode capital. While prior quantitative finance literature has separately explored recurrent temporal architectures [8], risk-sensitive reward objectives [4, 6], and dynamic walk-forward evaluation protocols [3], these techniques are rarely isolated under identical experimental conditions. When disparate studies use different datasets, transaction fee assumptions, feature definitions, and test periods, it becomes difficult to determine whether reported empirical differences arise from the neural architecture, the reward formulation, the turnover constraint, or the specifics of the data pipeline.

1.2 Research Objectives

To address this challenge through controlled experimentation, the specific objectives of this B.Tech. project are:

1. **Controlled Four-Stage Ablation:** To implement a single-factor ablation framework comprising four models—Model M1: PPO with a Multi-Layer Perceptron (MLP) baseline, Model M2: PPO with a recurrent Long Short-Term Memory (LSTM) encoder, Model M3: LSTM-PPO with recursive Differential Sharpe Ratio (DSR) reward, and Model M4: LSTM-PPO with DSR reward and an explicit quadratic weight-change penalty—to isolate the contributions of temporal memory, risk-adjusted reward shaping, and turnover control under identical data, cost, and training budgets.

2. **POMDP Trading Formulation:** To formulate daily portfolio allocation as a Partially Observable Markov Decision Process (POMDP), incorporating causal technical indicators, continuous Dirichlet policy sampling on the unit simplex, and exact transaction-cost accounting.

3. **Rigorous Chronological Evaluation:** To evaluate the four agents across an eight-year walk-forward test protocol (2018–2025) over four folds and three random seeds, comparing their risk-adjusted metrics, trading activity, and transaction costs against a passive Equal-Weight benchmark.

1.3 Research Question

This investigation focuses on answering one central research question:

> *"How do temporal memory, risk-aware reward shaping, and turnover regularization independently and cumulatively affect the risk-adjusted out-of-sample performance of a PPO-based daily trading agent?"*

From this question, four specific comparative aspects are examined:

- **RQ1 (Temporal Architecture):** How does an LSTM recurrent encoder alter out-of-sample risk-adjusted return and daily turnover compared with a static MLP baseline?
- **RQ2 (Risk-Aware Reward Shaping):** What is the empirical effect of replacing net return with the recursive online Differential Sharpe Ratio (DSR)?
- **RQ3 (Turnover Regularisation):** Does augmenting the reward with a quadratic weight-change penalty reduce turnover and transaction costs, and what is the trade-off with return?
- **RQ4 (Benchmark Competitiveness):** How do the learned active RL allocation policies compare against a passive, fully-invested Equal-Weight benchmark under identical transaction frictions?

Chapter 2 — Literature Review

The application of deep reinforcement learning to algorithmic trading and portfolio allocation has generated considerable academic interest. However, critical methodological challenges persist regarding cross-study comparability, temporal feature representations, reward formulation, action spaces, and out-of-sample evaluation.

In a comprehensive survey, Millea [4] documents that financial DRL studies frequently rely on heterogeneous asset classes, varied pre-processing steps, divergent trading friction assumptions, and non-standardised evaluation windows. These inconsistencies make it difficult to compare findings directly across the literature. Millea [4] also reviews risk-sensitive objective functions, emphasizing the Differential Sharpe Ratio (DSR), which computes an online, recursive estimate of risk-adjusted return by updating running first and second return moments, penalising variance rather than rewarding nominal return alone.

For policy optimisation in continuous environments, Proximal Policy Optimisation (PPO) [5] has become a standard actor-critic benchmark. By constraining policy updates via a clipped probability ratio objective, PPO ensures stable monotonic improvement without the hyperparameter fragility and sample variance characteristic of standard policy gradient algorithms.

In the domain of temporal modeling, financial time series exhibit momentum, autocorrelation, and cyclical regime shifts that cannot be discerned from an isolated single-day snapshot. Long Short-Term Memory (LSTM) recurrent networks [1] provide gating mechanisms that maintain memory states over sequential observations, alleviating vanishing gradients. Investigating temporal architectures in trading, Zou et al. [8] implement a cascaded LSTM-PPO network for equity trading, demonstrating that multi-day historical windows help capture sequential price dependencies. Their experiments indicate that a lookback window of $W = 30$ trading days paired with an LSTM hidden dimension of $H = 512$ provides an effective temporal summary of price dynamics. However, their policy rewards remain based on raw return, leaving the interaction between recurrent feature representations and risk-adjusted reward shaping unexamined.

Regarding reward formulation, Huang et al. [2] design a self-rewarding reinforcement learning framework combining expert-crafted trading rules with a learned internal reward network. While

their findings highlight the importance of reward engineering, their approach relies on auxiliary neural components rather than analytically derived risk metrics such as DSR. Addressing market volatility, Wang and Liu [6] propose an adaptive risk-sensitive reinforcement learning framework for portfolio optimization, demonstrating downside protection across shifting conditions; however, their setup alters the neural backbone, reward formulation, and execution constraints simultaneously, preventing single-factor attribution.

For empirical evaluation rigor, Liu et al. [3] introduce dynamic financial environments and a rolling train–validation–test framework that continuously advances training, validation, and testing windows chronologically. This setup addresses survivorship bias and captures macroeconomic regime shifts, elevating evaluation standards in algorithmic finance.

Finally, action space parameterisation is a fundamental mathematical consideration in portfolio management. The portfolio weight vector must strictly satisfy the long-only unit simplex constraints: $w_i \geq 0$ and $\sum_{i=1}^{M} w_i = 1$. While earlier studies commonly applied Softmax activation to unconstrained Gaussian outputs, Yang et al. [7] demonstrate that parameterising the actor network over a continuous multivariate Dirichlet distribution natively enforces simplex constraints on the probability manifold without requiring heuristic clipping or distorted probability densities.

Chapter 3 — Research Gap and Problem Formulation

3.1 Identified Research Gap

While significant progress has been made across temporal modeling, risk-sensitive rewards, and walk-forward validation protocols, existing studies have typically evaluated these mechanisms within isolated frameworks using divergent market universes, varied cost assumptions, and concurrent architectural changes.

Based on the literature reviewed, there remains a clear need for controlled empirical comparisons that systematically isolate the contributions of temporal memory, risk-aware rewards, and turnover regularization under a common, realistic experimental setup. Without holding the asset universe, feature set, transaction-cost model, training budget, and chronological evaluation protocol constant, it is difficult to determine whether reported differences in out-of-sample performance stem from the sequence encoder, the reward function, turnover control, or differences in the experimental setup itself.

Table 3.1 positions this thesis relative to the reviewed literature, highlighting the controlled single-factor ablation design adopted here.

Table 3.1: Positioning of this work relative to the reviewed literature.

Paper	Temporal Memory	Risk-Aware / Reward	Turnover Control	Continuous Allocation	Controlled Ablation
Millea (2021) [4]		✓ *(risk-aware rewards discussed)*			
Zou et al. (2024) [8]	✓				
Huang et al. (2024) [2]		✓			
Liu et al. (2024) [3]				✓ *(portfolio weights)*	
Wang & Liu (2025) [6]		✓ *(risk-sensitive reward)*	✓	✓	
Yang et al. (2022) [7]				✓ *(Dirichlet distribution)*	
This work	✓	✓ *(DSR)*	✓ *(turnover penalty)*	✓	✓

3.2 POMDP Formulation of Daily Portfolio Allocation

We formulate daily portfolio allocation as a Partially Observable Markov Decision Process (POMDP), defined by the 7-tuple:

$$\mathcal{P} = \langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \Omega, \mathcal{O}, \gamma \rangle, \tag{3.1}$$

where $\mathcal{S}$ is the unobservable state space, $\mathcal{A}$ is the action space, $\mathcal{T}$ represents transition dynamics, $\mathcal{R}$ is the reward function, Ω is the observation space, $\mathcal{O}$ is the observation probability distribution, and $\gamma \in [0, 1)$ is the discount factor.

3.2.1 Why Daily Financial Trading is a POMDP

In a fully observable Markov Decision Process (MDP), the Markov property requires that future state transitions depend solely on the current state and action: $\mathbb{P}(\mathbf{s}_{t+1} \mid \mathbf{s}_t, \mathbf{a}_t) = \mathbb{P}(\mathbf{s}_{t+1} \mid \mathbf{s}_t, \ldots, \mathbf{s}_0)$. In financial trading, this assumption does not hold when applied directly to observable market indicators:

1. **Unobservable True Market State $\mathcal{S}$:** The complete economic state of financial markets includes macroeconomic fundamentals, central bank expectations, institutional limit order book queues, private liquidity flows, and participant psychology. These variables are latent and cannot be directly observed by an agent trading from public daily price feeds.

2. **Partial and Noisy Observations Ω:** The agent receives only an observation vector $\mathbf{o}_t \in \Omega$ drawn from the observation model $\mathcal{O}(\mathbf{o}_t \mid \mathbf{s}_t)$, consisting of daily open, high, low, close prices, trading volumes, and derived technical indicators.

3. **Non-Markovian Indicator Dynamics:** An identical single-day technical observation vector (e.g., RSI = 65, MACD crossover) can result in completely different future price movements depending on latent market conditions (such as the beginning of a sustained trend versus an institutional distribution phase).

Because single-day observations $\mathbf{o}_t$ do not satisfy the Markov property, an agent benefits from maintaining an internal temporal representation or belief state $b(\mathbf{s}_t)$ over historical observation sequences $\mathbf{F}_t = [\mathbf{s}_{t-W+1}, \ldots, \mathbf{s}_t]$. In this work, recurrent neural networks (LSTM) are used to construct this temporal summary $\mathbf{h}_t \approx b(\mathbf{s}_t)$ end-to-end, directly mitigating the effects of partial observability.

3.2.2 Action Space $\mathcal{A}$ and Simplex Manifold

The portfolio consists of $N = 10$ risky equities and 1 cash account, yielding an action dimension of $M = N + 1 = 11$. The action vector $\mathbf{a}_t = [a_{t,1}, \ldots, a_{t,N+1}]^\top$ represents target portfolio weights, strictly constrained to lie on the unit simplex:

$$\mathcal{A} \equiv \Delta^N = \left\{ \mathbf{w}_t \in \mathbb{R}^{N+1} \;\middle|\; w_{i,t} \geq 0 \; \forall i, \quad \sum_{i=1}^{N} w_{i,t} + w_{\text{cash},t} = 1 \right\}. \tag{3.2}$$

Following Yang et al. [7], this constraint is enforced directly by parameterising the actor network over a multivariate Dirichlet distribution, eliminating the need for heuristic projection or unconstrained Gaussian clipping.

Chapter 4 — Dataset and Experimental Methodology

4.1 Equity Universe and Data Acquisition

The experimental stock universe comprises $N = 10$ prominent large-cap equities actively traded on the National Stock Exchange of India (NSE), spanning diverse economic sectors:

1. **RELIANCE.NS** (Energy and Petrochemicals / Conglomerate)
2. **TCS.NS** (Information Technology Services)
3. **HDFCBANK.NS** (Private Commercial Banking)
4. **ICICIBANK.NS** (Private Commercial Banking)
5. **ITC.NS** (Consumer Goods and Diversified Conglomerate)
6. **HINDUNILVR.NS** (Fast-Moving Consumer Goods)
7. **LT.NS** (Engineering and Infrastructure Construction)
8. **SUNPHARMA.NS** (Pharmaceuticals and Healthcare)
9. **MARUTI.NS** (Automotive Manufacturing)
10. **BHARTIARTL.NS** (Telecommunications and Digital Services)

In addition to these ten equities, the portfolio includes an explicit cash asset (Asset 11), providing the agent with the operational flexibility to hold liquidity and de-risk during volatile market phases.

Daily price and volume records were acquired from Yahoo Finance via the `yfinance` API spanning 1 January 2009 to 31 December 2025. Data acquisition utilized `auto_adjust=True`: price series (Open, High, Low, and Close) are backward-adjusted for both stock splits and cash dividends, whereas trading volume is adjusted for splits. Data from 2009–2010 serves as an initial burn-in buffer to initialize 200-day moving averages, ensuring that all technical indicators are fully settled by 1 January 2011. The common trading calendar comprises 3,699 active trading days.

4.2 Feature Engineering and Market Indicators

For each asset $i \in \{1, \ldots, N\}$ on trading day t, a compact feature vector $\mathbf{f}_{i,t} \in \mathbb{R}^{13}$ is computed using causal market and technical price transformations. To ensure cross-asset comparability, features are normalized by prevailing prices or rolling averages:

1. **Daily Log Return ($f_{1,i,t}$):** $r_{i,t}^{\log} = \ln(C_{i,t}/C_{i,t-1})$.
2. **Close-to-Open Relative Return ($f_{2,i,t}$):** $o_{i,t}^{\text{rel}} = C_{i,t}/O_{i,t} - 1$.
3. **High-to-Low Relative Spread ($f_{3,i,t}$):** $hl_{i,t}^{\text{rel}} = H_{i,t}/L_{i,t} - 1$.
4. **Normalized Relative Volume ($f_{4,i,t}$):** $v_{i,t}^{\text{rel}} = \ln(V_{i,t}/\text{SMA}_{20}(V_{i,t}))$.
5. **50-day EMA Relative Deviation ($f_{5,i,t}$):** $ema_{50,i,t}^{\text{rel}} = C_{i,t}/\text{EMA}_{50}(C_{i,t}) - 1$, with smoothing multiplier $\alpha_n = \frac{2}{n+1}$.
6. **200-day EMA Relative Deviation ($f_{6,i,t}$):** $ema_{200,i,t}^{\text{rel}} = C_{i,t}/\text{EMA}_{200}(C_{i,t}) - 1$.
7. **Relative Strength Index ($f_{7,i,t}$):** Wilder's 14-period RSI centered around zero:

$$\text{RS}_{i,t} = \frac{\bar{U}_{i,t}}{\bar{D}_{i,t} + 10^{-8}}, \quad \text{RSI}_{i,t} = 100 - \frac{100}{1 + \text{RS}_{i,t}}, \quad f_{7,i,t} = \frac{\text{RSI}_{i,t}}{100} - 0.5. \tag{4.1}$$

8. **Moving Average Convergence Divergence ($f_{8,i,t}$):** Normalized trend divergence:

$$\mathrm{MACD}_{i,t} = \frac{\mathrm{EMA}_{12}(C_{i,t}) - \mathrm{EMA}_{26}(C_{i,t})}{C_{i,t}}. \tag{4.2}$$

9. **MACD Signal Line ($f_{9,i,t}$):** $\mathrm{Signal}_{i,t} = \mathrm{EMA}_9(\mathrm{MACD}_{i,t})/C_{i,t}$.

10. **Average True Range ($f_{10,i,t}$):** Normalized price volatility:

$$\mathrm{ATR}^{\mathrm{rel}}_{i,t} = \frac{\mathrm{SMA}_{14}(\mathrm{TR}_{i,t})}{C_{i,t}}, \quad \text{where } \mathrm{TR}_t = \max(H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|). \tag{4.3}$$

11. **Bollinger Band %B ($f_{11,i,t}$):** Position relative to 2-standard-deviation envelopes:

$$\%B_{i,t} = \frac{C_{i,t} - \mathrm{LB}_{i,t}}{\mathrm{UB}_{i,t} - \mathrm{LB}_{i,t} + 10^{-8}}. \tag{4.4}$$

12. **Bollinger Bandwidth ($f_{12,i,t}$):** Volatility width: $\mathrm{BW}_{i,t} = (\mathrm{UB}_{i,t} - \mathrm{LB}_{i,t})/\mathrm{MB}_{i,t}$.

13. **Average Directional Index ($f_{13,i,t}$):** Wilder's 14-period normalized trend: $\mathrm{ADX}^{\mathrm{norm}}_{i,t} = \mathrm{ADX}_{14,i,t}/100$.

4.3 State Representation and Temporal Sequences

The instantaneous observation vector concatenates the 13 features across all $N = 10$ equities (130 market features) with the current portfolio weight vector $\mathbf{w}^{\mathrm{cur}}_t \in \mathbb{R}^{N+1}$ (10 equities plus cash, 11 weights):

$$\mathbf{x}_t = \mathrm{concat}\,(\mathbf{f}_{1,t}, \ldots, \mathbf{f}_{N,t}) \in \mathbb{R}^{130}, \tag{4.5}$$

$$\mathbf{s}_t = \left[\mathbf{x}_t^\top,\ (\mathbf{w}^{\mathrm{cur}}_t)^\top\right]^\top \in \mathbb{R}^{141}. \tag{4.6}$$

The total daily state dimension is $D_{\mathrm{state}} = 141$. For models using recurrent memory (M2–M4), a sliding historical window of $W = 30$ trading days is maintained:

$$\mathbf{F}_t = [\mathbf{s}_{t-W+1}, \ldots, \mathbf{s}_t]^\top \in \mathbb{R}^{30\times 141}. \tag{4.7}$$

Feature standardization is fitted strictly on the training partition of each fold using mean $\boldsymbol{\mu}_{\mathrm{train}}$ and standard deviation $\boldsymbol{\sigma}_{\mathrm{train}}$, with standardized values clipped to $[-5.0, 5.0]$. All features use causal backward operators, preventing look-ahead leakage.

4.4 Walk-Forward Evaluation Protocol

Following Liu et al. [3], models are trained and tested using an 8-year rolling-origin walk-forward protocol across four folds. Each fold consists of 6 years of training data, 1 year of validation data, and 2 years of held-out out-of-sample test data, advancing by 2 years per fold (Table 4.1 and Figure 4.1). The test periods are non-overlapping and span 2018 through 2025. It is important to distinguish the number of historical trading days in the dataset splits ($\approx 1,478$ training days and ≈ 490 test days) from the reinforcement learning training budget, which is fixed at 300,000 environment interaction steps across all models.

To assess statistical variability, each configuration is trained across three fixed random seeds ($S \in \{0, 1, 2\}$). During training, validation Sharpe ratio is evaluated periodically every 25,000 environment steps, and the checkpoint achieving the highest validation Sharpe ratio is saved. At the

Table 4.1: Chronological specification of the four walk-forward folds.

Fold	Training Period (6y)	Validation Period (1y)	Test Period (2y)	Training Trading Days	Test Trading Days
1	2011-01-01 to 2016-12-31	2017-01-01 to 2017-12-31	2018-01-01 to 2019-12-31	1,478	489
2	2013-01-01 to 2018-12-31	2019-01-01 to 2019-12-31	2020-01-01 to 2021-12-31	1,475	496
3	2015-01-01 to 2020-12-31	2021-01-01 to 2021-12-31	2022-01-01 to 2023-12-31	1,476	495
4	2017-01-01 to 2022-12-31	2023-01-01 to 2023-12-31	2024-01-01 to 2025-12-31	1,479	494

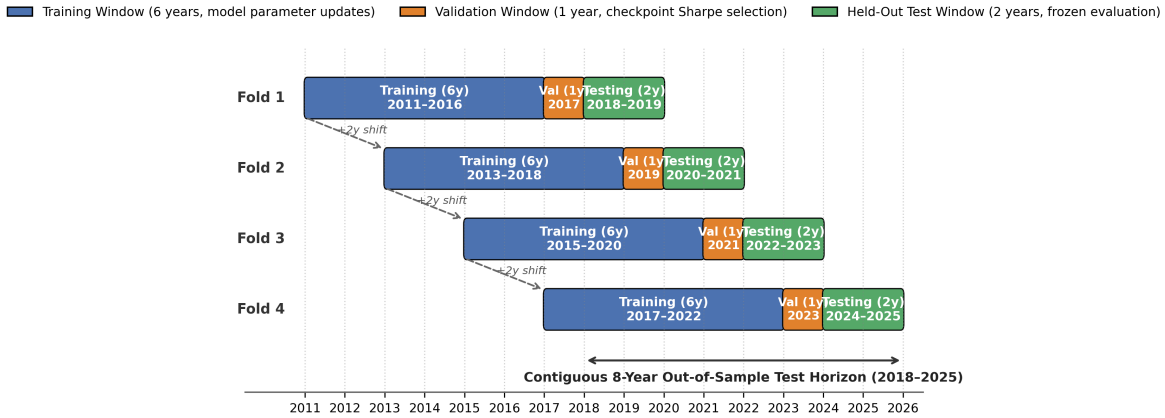


Figure 4.1: Chronological four-fold rolling walk-forward evaluation protocol with 6-year training, 1-year validation, and 2-year testing windows, advancing by 2 years per fold (adapted from Liu et al. (2024), Figure 5).

conclusion of 300,000 steps, this best-validation checkpoint is restored and evaluated once on the out-of-sample test window.

Methodological Disclosure: We note that earlier exploratory runs of Model M1 were conducted on Fold 1 test data during preliminary system debugging. We distinguish those preliminary debugging iterations from the final reported runs; test-set feedback was not used to adjust hyperparameters or select architectures.

Chapter 5 — Model Architecture and Mathematical Formulation

5.1 Overview of the Actor–Critic Architecture

The portfolio management agents are trained using Proximal Policy Optimisation (PPO) [5] formulated within an Actor–Critic framework. The system architecture decouples state representation learning, continuous allocation policy parameterisation on the simplex manifold, and state-value baseline estimation:

1. An **Encoder** processes market observations to generate a compact 512-dimensional latent representation $\mathbf{h}_t \in \mathbb{R}^{512}$.
2. An **Actor head** maps $\mathbf{h}_t$ to the concentration parameters of a multivariate Dirichlet distribution that outputs portfolio allocations across $M = 11$ assets (10 equities plus cash).
3. A **Critic head** maps $\mathbf{h}_t$ to a scalar baseline estimate $V_\phi(\mathbf{s}_t) \in \mathbb{R}$ approximating expected cumulative discounted future reward.

Figure 5.1 illustrates the information flow from input tensor to executed portfolio weights, and Figure 5.2 portrays the rollout interaction and PPO training loop.

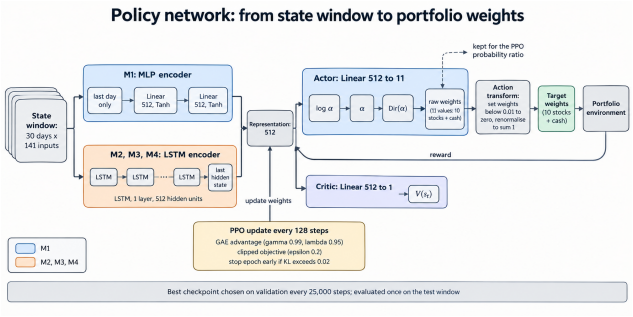


Figure 5.1: Policy architecture from state to portfolio weights.

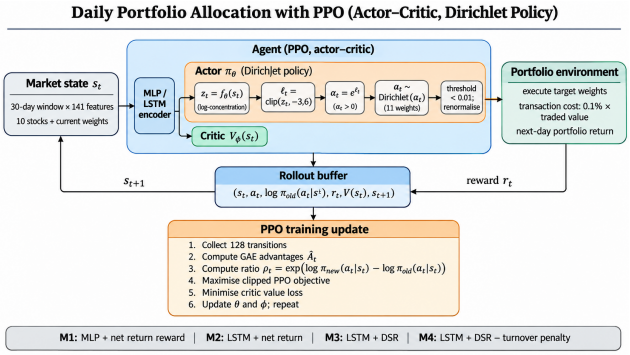


Figure 5.2: PPO Dirichlet allocation and environment loop.

5.2 Encoder Formulations: MLP versus LSTM

To evaluate the empirical utility of temporal memory, two alternative encoder backbones are examined:

5.2.1 Multi-Layer Perceptron (MLP) Encoder (Model M1)

The feedforward MLP baseline observes solely the instantaneous observation vector at day t, $\mathbf{s}_t \in \mathbb{R}^{141}$, discarding multi-period historical trajectories. The representation $\mathbf{h}_t$ and raw actor output $\mathbf{z}_t$ are computed via two fully-connected layers with Tanh activations:

$$\mathbf{h}_t^{(1)} = \tanh\left(\mathbf{W}_1\mathbf{s}_t + \mathbf{b}_1\right), \quad \mathbf{W}_1 \in \mathbb{R}^{512\times 141},\ \mathbf{b}_1 \in \mathbb{R}^{512}, \tag{5.1}$$

$$\mathbf{h}_t = \tanh\left(\mathbf{W}_2\mathbf{h}_t^{(1)} + \mathbf{b}_2\right), \quad \mathbf{W}_2 \in \mathbb{R}^{512\times 512},\ \mathbf{b}_2 \in \mathbb{R}^{512}, \tag{5.2}$$

$$\mathbf{z}_t = f_\theta^{\text{MLP}}(\mathbf{s}_t) = \mathbf{W}_a\mathbf{h}_t + \mathbf{b}_a, \quad \mathbf{W}_a \in \mathbb{R}^{11\times 512},\ \mathbf{b}_a \in \mathbb{R}^{11}. \tag{5.3}$$

5.2.2 Recurrent LSTM Encoder (Models M2–M4)

To capture sequential momentum and multi-period dependencies, Models M2, M3, and M4 pass the sliding history window $\mathbf{F}_t = [\mathbf{s}_{t-W+1}, \ldots, \mathbf{s}_t] \in \mathbb{R}^{30\times 141}$ ($W = 30$ trading days) through a recurrent Long Short-Term Memory (LSTM) network [1, 8]:

$$(\mathbf{h}_t, \mathbf{c}_t) = \text{LSTM}_\psi(\mathbf{F}_t), \tag{5.4}$$

$$\mathbf{z}_t = f_\theta^{\text{LSTM}}(\mathbf{F}_t) = \mathbf{W}_a\mathbf{h}_t + \mathbf{b}_a, \tag{5.5}$$

where ψ denotes recurrent weights and $\mathbf{h}_t \in \mathbb{R}^{512}$ is the final hidden state across the 30 daily steps. The LSTM uses internal forget gates (to discard stale price patterns), input gates (to integrate new market signals), and output gates (to modulate the emitted hidden representation $\mathbf{h}_t$). Rather than acting as a directional price forecaster, $\mathbf{h}_t$ serves as a learned temporal summary for sequential allocation.

5.3 Continuous Dirichlet Policy on the Simplex

Portfolio allocations must strictly satisfy the long-only unit simplex constraints:

$$w_{i,t}^{\text{target}} \geq 0, \quad w_{\text{cash},t}^{\text{target}} \geq 0, \quad \sum_{i=1}^{N} w_{i,t}^{\text{target}} + w_{\text{cash},t}^{\text{target}} = 1. \tag{5.6}$$

Following Yang et al. [7], the continuous policy is parameterised using a multivariate Dirichlet distribution.

5.3.1 Actor Mapping to Dirichlet Parameters

The actor head outputs unconstrained real values $\mathbf{z}_t \in \mathbb{R}^{11}$. Concentration parameters must be strictly positive ($\alpha_i > 0$). In our implementation, $\mathbf{z}_t$ represents log-concentration values, which are clamped to a numerically safe range and exponentiated:

$$\boldsymbol{\ell}_t = \text{clip}\left(\mathbf{z}_t, -3.0, 6.0\right), \tag{5.7}$$

$$\alpha_{i,t} = \exp\left(\ell_{i,t}\right) \in \left[e^{-3}, e^{6}\right] \approx [0.0498, 403.43]. \tag{5.8}$$

Clamping prevents underflow (avoiding degenerate densities at simplex vertices) and overflow (avoiding gradient instability in the Gamma function derivative).

5.3.2 Action Sampling, Thresholding, and Log-Probabilities

It is essential to distinguish between the three operational forms of the action:

1. **Raw Sampled Action ($\mathbf{a}_t$):** During environment rollout, a continuous action vector $\mathbf{a}_t \in \Delta^N$ is sampled from the Dirichlet probability density function:

$$\pi_\theta(\mathbf{a}_t \mid \mathbf{h}_t) = \frac{\Gamma\left(\sum_{i=1}^{N+1} \alpha_{i,t}\right)}{\prod_{i=1}^{N+1} \Gamma(\alpha_{i,t})} \prod_{i=1}^{N+1} a_{i,t}^{\alpha_{i,t}-1}, \tag{5.9}$$

   via normalized Gamma variates: $g_{i,t} \sim \text{Gamma}(\alpha_{i,t}, 1)$ with $a_{i,t} = g_{i,t} / \sum_j g_{j,t}$. The expected portfolio allocation is $\mathbb{E}[a_{i,t}] = \alpha_{i,t} / \sum_j \alpha_{j,t}$, which is executed directly during deterministic evaluation.

2. **Thresholded Target Weights ($\mathbf{w}_t^{\text{target}}$):** Because a Dirichlet sample with positive parameters assigns non-zero mass across all assets, tiny positions can induce unnecessary transaction fees. A threshold $\tau = 0.01$ (1%) is applied, zeroing minuscule allocations and renormalising survivors:

$$\tilde{w}_{i,t} = \begin{cases} a_{i,t}, & \text{if } a_{i,t} \geq \tau, \\ 0, & \text{if } a_{i,t} < \tau, \end{cases} \qquad w_{i,t}^{\text{target}} = \frac{\tilde{w}_{i,t}}{\sum_{j=1}^{N+1} \tilde{w}_{j,t}}. \tag{5.10}$$

3. **Log Probability Evaluation for PPO:** For PPO importance sampling, the log-probability is evaluated strictly on the *raw continuous action* $\mathbf{a}_t$ under the Dirichlet density:

$$\log \pi_\theta(\mathbf{a}_t \mid \mathbf{h}_t) = \ln \Gamma\left(\sum_{i=1}^{N+1} \alpha_{i,t}\right) - \sum_{i=1}^{N+1} \ln \Gamma(\alpha_{i,t}) + \sum_{i=1}^{N+1} (\alpha_{i,t} - 1) \ln a_{i,t}. \tag{5.11}$$

   The raw action $\mathbf{a}_t$ (and not $\mathbf{w}_t^{\text{target}}$) is stored in the rollout buffer to preserve the validity of probability density evaluations.

5.4 Portfolio Accounting and Transaction Timing

To ensure causality and prevent look-ahead bias, the transaction timeline operates as follows:

1. At day t, the agent observes historical market information $\mathbf{s}_t$ (or window $\mathbf{F}_t$) available up to market close.
2. The agent determines target portfolio weights $\mathbf{w}_t^{\text{target}}$ for day t.
3. The portfolio is rebalanced from current weights $\mathbf{w}_t^{\text{cur}}$ to $\mathbf{w}_t^{\text{target}}$, incurring transaction cost C_t.

4. The rebalanced portfolio earns the subsequent asset returns from day t to day $t+1$:

$$R_{i,t+1} = \frac{C_{i,t+1}}{C_{i,t}} - 1. \tag{5.12}$$

Rebalancing the N risky equities incurs a proportional fee $c_{\text{trans}} = 0.001$ (10 bps), while reallocations to or from the cash leg incur no fee:

$$\text{Turnover}_t = \sum_{i=1}^{N} \left| w_{i,t}^{\text{target}} - w_{i,t}^{\text{cur}} \right|, \tag{5.13}$$

$$C_t = c_{\text{trans}} V_t \cdot \text{Turnover}_t. \tag{5.14}$$

As a simplifying modeling assumption, cash earns zero nominal interest (multiplier 1.0). End-of-day wealth V_{t+1} evolves as:

$$V_{t+1} = (V_t - C_t) \left[\sum_{i=1}^{N} w_{i,t}^{\text{target}} (1 + R_{i,t+1}) + w_{\text{cash},t}^{\text{target}} \right]. \tag{5.15}$$

Transaction cost C_t is deducted once from capital prior to return compounding. The realized net rate of portfolio return is:

$$R_{\text{net},t} = \frac{V_{t+1} - V_t}{V_t}. \tag{5.16}$$

Entering day $t+1$, asset weights naturally drift to $w_{i,t+1}^{\text{cur}} = w_{i,t}^{\text{target}}(1 + R_{i,t+1})/G_t$ $(i \leq N)$ and $w_{\text{cash},t+1}^{\text{cur}} = w_{\text{cash},t}^{\text{target}}/G_t$, where $G_t = V_{t+1}/(V_t - C_t)$ is the gross growth multiplier.

5.5 Model Reward Formulations (M1 to M4)

The reward signal r_t emitted to the agent defines the ablation dimension:

1. **Models M1 and M2 (Net Return Reward):**

$$r_t = R_{\text{net},t}. \tag{5.17}$$

2. **Model M3 (Differential Sharpe Ratio):** Following Millea [4], first and second return moments ($A_t \approx \mathbb{E}[R]$ and $B_t \approx \mathbb{E}[R^2]$) update recursively with adaptation rate $\eta = 0.01$:

$$A_t = A_{t-1} + \eta(R_{\text{net},t} - A_{t-1}), \tag{5.18}$$

$$B_t = B_{t-1} + \eta(R_{\text{net},t}^2 - B_{t-1}). \tag{5.19}$$

   With moment changes $\Delta A_t = A_t - A_{t-1}$ and $\Delta B_t = B_t - B_{t-1}$, the online DSR reward is:

$$D_t = \frac{B_{t-1}\Delta A_t - \frac{1}{2}A_{t-1}\Delta B_t}{\left(B_{t-1} - A_{t-1}^2 + \varepsilon_{\text{DSR}}\right)^{3/2}}, \tag{5.20}$$

   where $\varepsilon_{\text{DSR}} = 10^{-8}$ prevents numerical division by zero. For Model M3, $r_t = D_t$.

3. **Model M4 (DSR + Quadratic Penalty):** An explicit quadratic penalty on risky asset rebalancing is subtracted from the DSR reward:

$$r_t = D_t - \lambda_{\text{turn}} \sum_{i=1}^{N} \left(w_{i,t}^{\text{target}} - w_{i,t}^{\text{cur}} \right)^2, \tag{5.21}$$

with $\lambda_{\text{turn}} = 0.1$ selected via preliminary validation tuning. Note that this quadratic penalty is an optimization regularizer on the reward; it is distinct from the L1 turnover metric and is not deducted from portfolio wealth V_t.

5.6 PPO Policy Optimisation

GAE computes advantage estimates $\hat{A}_t$ using temporal-difference residuals $\delta_t = \tilde{r}_t + \gamma V_\phi(\mathbf{s}_{t+1}) - V_\phi(\mathbf{s}_t)$ with discount $\gamma = 0.99$ and trace decay $\lambda_{\text{GAE}} = 0.95$:

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma \lambda_{\text{GAE}})^l \delta_{t+l}. \tag{5.22}$$

The probability ratio is evaluated for the identical raw action $\mathbf{a}_t$:

$$\rho_t(\theta) = \frac{\pi_\theta(\mathbf{a}_t \mid \mathbf{h}_t)}{\pi_{\theta_{\text{old}}}(\mathbf{a}_t \mid \mathbf{h}_t)} = \exp\left(\log \pi_\theta(\mathbf{a}_t \mid \mathbf{h}_t) - \log \pi_{\theta_{\text{old}}}(\mathbf{a}_t \mid \mathbf{h}_t)\right). \tag{5.23}$$

Policy parameters θ and critic parameters ϕ are optimized using Adam by minimising the joint loss [5]:

$$\mathcal{L}(\theta, \phi) = -\hat{\mathbb{E}}_t \left[\min\left(\rho_t(\theta)\hat{A}_t,\ \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \right) \right] + c_1 \hat{\mathbb{E}}_t \left[(V_\phi(\mathbf{s}_t) - \hat{V}_t^{\text{target}})^2 \right] - c_2 \hat{\mathbb{E}}_t[\mathcal{H}(\pi_\theta)], \tag{5.24}$$

with clipping range $\epsilon = 0.2$, critic coefficient $c_1 = 0.5$, and analytical Dirichlet entropy bonus coefficient $c_2 = 0.01$.

Chapter 6 — Controlled Ablation Study Design

6.1 Methodological Rationale for Controlled Ablation

In deep reinforcement learning research, empirical comparisons often suffer from reproducibility crises because novel models introduce multiple architectural alterations, revised reward functions, and distinct hyperparameter tunings simultaneously. Under such confounded experimental designs, it is impossible to determine whether empirical gains arise from neural representational power, risk-adjusted reward shaping, or hyperparameter coincidences.

To establish rigorous scientific causality, this research adopts a strict *single-factor controlled ablation study*. Four model configurations (M1 through M4) are constructed such that each consecutive transition alters exactly one algorithmic mechanism while holding all other parameters, random seeds, training budgets, network widths, transaction costs, and data folds strictly invariant.

6.2 Specification of Ablation Models M1–M4

The four ablation models are defined as follows:

1. **Model M1 (MLP Baseline + Net Return):** Evaluates a two-layer feedforward MLP observing solely the instantaneous market observation $\mathbf{s}_t \in \mathbb{R}^{141}$. The reward is the realised net rate of return ($r_t = R_{\text{net},t}$) without turnover regularisation ($\lambda_{\text{turn}} = 0$). This serves as the static, non-temporal baseline.

2. **Model M2 (LSTM Recurrent Encoder + Net Return):** Replaces the static MLP with a single-layer LSTM network with hidden dimension $H = 512$, processing the sliding 30-day historical window. The reward remains net return ($r_t = R_{\text{net},t}$). Transition M1 $\rightarrow$ M2 strictly isolates the impact of **temporal memory and sequence modeling**.

Algorithm 1: PPO Daily Portfolio Allocation with Dirichlet Policy

Input: Market series $\mathbf{X}$, returns $\mathbf{R}$, cost c_{trans}, threshold τ, penalty λ_{turn}, PPO parameters $\gamma, \lambda_{\text{GAE}}, \epsilon, \eta_{\text{LR}}$.
Initialisation: Actor θ, Critic ϕ, reward normaliser $\mathcal{N}$, initial weights $\mathbf{w}_0^{\text{cur}} \leftarrow [0, \ldots, 0, 1]^\top$ (cash), DSR moments $A \leftarrow 0, B \leftarrow 0$.

for iteration $= 1, 2, \ldots$ until $T_{\text{total}} = 300,000$ steps **do**
  Reset rollout buffer $\mathcal{B} \leftarrow \emptyset$.
  for $t = 1$ to $T_{\text{rollout}} = 128$ **do**
    Compute representation $\mathbf{h}_t \leftarrow \text{Encoder}(\mathbf{s}_t)$ (MLP for M1; LSTM for M2–M4).
    Compute concentrations $\boldsymbol{\ell}_t \leftarrow \text{clip}(\mathbf{W}_a \mathbf{h}_t + \mathbf{b}_a, -3, 6)$ and $\boldsymbol{\alpha}_t \leftarrow \exp(\boldsymbol{\ell}_t)$.
    Sample raw action variates $g_i \sim \text{Gamma}(\alpha_{i,t}, 1)$, set $\mathbf{a}_t \leftarrow \mathbf{g} / \sum_i g_i$.
    Evaluate old log-likelihood $lp_t \leftarrow \log \pi_\theta(\mathbf{a}_t \mid \mathbf{h}_t)$ and value $V_t \leftarrow V_\phi(\mathbf{h}_t)$.
    Apply threshold $\tau = 0.01$: $\tilde{w}_i \leftarrow a_{i,t} \cdot \mathbf{1}_{a_{i,t} \geq \tau}$, set $\mathbf{w}_t^{\text{target}} \leftarrow \tilde{\mathbf{w}} / \sum_j \tilde{w}_j$.
    Execute trade: compute Turnover_t, cost C_t, net return $R_{\text{net},t}$, new wealth V_{t+1}.
    Compute model reward r_t (Net return for M1/M2; DSR for M3; DSR $-\lambda_{\text{turn}} \sum (\Delta w_i)^2$ for M4).
    Store transition $(\mathbf{s}_t, \mathbf{a}_t, lp_t, V_t, r_t)$ in buffer $\mathcal{B}$.
  end for
  Compute GAE advantages $\hat{A}_t$ and return targets $\hat{V}_t^{\text{target}}$.
  for epoch $= 1$ to $K = 10$ **do**
    for each minibatch of size $B = 64$ **do**
      Evaluate current log-likelihoods $\log \pi_\theta(\mathbf{a}_i \mid \mathbf{h}_i)$ and values $V_\phi(\mathbf{h}_i)$.
      Compute probability ratio $\rho_i(\theta) \leftarrow \exp(\log \pi_\theta - lp_i)$ and multi-task loss $\mathcal{L}(\theta, \phi)$.
      Update (θ, ϕ) via Adam optimiser ($\eta_{\text{LR}} = 10^{-4}$) with gradient clipping 0.5.
    end for
  end for
end for

Figure 5.3: Complete training and policy update algorithm for PPO portfolio allocation.

3. **Model M3 (LSTM Encoder + Differential Sharpe Ratio):** Retains the identical LSTM backbone ($H = 512$), but replaces the net return reward with the recursive online Differential Sharpe Ratio ($r_t = D_t$). Transition M2 $\rightarrow$ M3 strictly isolates the impact of **risk-aware reward engineering** against raw return maximisation.

4. **Model M4 (LSTM + DSR + Quadratic Weight-Change Regularisation):** Retains the LSTM backbone and DSR reward, augmented by an explicit quadratic penalty on risky asset rebalancing ($r_t = D_t - \lambda_{\text{turn}} \sum_{j=1}^{N} (w_{j,t}^{\text{target}} - w_{j,t}^{\text{cur}})^2$) with $\lambda_{\text{turn}} = 0.1$. Transition M3 $\rightarrow$ M4 strictly isolates the impact of **explicit turnover regularisation and transaction-friction suppression**.

Table 6.1 summarises the architectural and functional properties of the four configurations.

Table 6.1: Systematic ablation matrix across models M1 to M4.

Configuration Property	**Model M1**	**Model M2**	**Model M3**	**Model M4**
Encoder Backbone	MLP (2-layer)	LSTM (1-layer)	LSTM (1-layer)	LSTM (1-layer)
Temporal Window (W)	1 day (last row)	30 days	30 days	30 days
Latent Dimension (H)	512	512	512	512
Policy Head	Dirichlet Δ^{11}	Dirichlet Δ^{11}	Dirichlet Δ^{11}	Dirichlet Δ^{11}
Action Threshold (τ)	0.01	0.01	0.01	0.01
Reward Formulation	Net Return	Net Return	DSR (D_t)	DSR (D_t)
Turnover Regularisation	None ($\lambda = 0$)	None ($\lambda = 0$)	None ($\lambda = 0$)	Quadratic Penalty ($\lambda = 0.1$)
Environment Cost Model	0.1% per turnover	0.1% per turnover	0.1% per turnover	0.1% per turnover
Training Budget	300,000 steps	300,000 steps	300,000 steps	300,000 steps
Ablation Mechanism	*Baseline*	**+ Temporal**	**+ Risk-Aware**	**+ Turnover Penalty**

6.3 Control Invariants and Verification

To guarantee that no hidden variance compromises the ablation comparisons:

1. **Weight Initialisation Parity:** In `src/models.py`, the linear Actor head ($\mathbf{W}_a, \mathbf{b}_a$) and Critic head ($\mathbf{W}_c, \mathbf{b}_c$) are instantiated *before* the encoder network. When initialised under an identical pseudo-random seed, models M1 and M2 possess bitwise identical initial weights in their policy and value heads.

2. **Equal Computational Budget:** All models are trained for exactly 300,000 environment interaction steps with an identical Adam learning rate (10^{-4}), rollout horizon (128), and minibatch size (64).

3. **Identical Market Realisation and Costs:** Each model experiences the exact same price sequences, transaction cost deductions (0.1%), and execution mechanics.

4. **Automated Assertion:** The execution harness enforces `assert_controlled_ablation()` at startup, raising an assertion failure if any model transition modifies more than exactly one algorithmic dimension.

Chapter 7 — Results and Discussion

7.1 Overall Out-of-Sample Performance

Out-of-sample evaluation was conducted across four chronological 2-year test windows spanning 2018 through 2025. Table 7.1 reports the pooled test performance metrics (averaged across four folds and three random seeds) for Models M1–M4 alongside the passive Equal-Weight (EW) benchmark.

The primary empirical finding is that the passive Equal-Weight benchmark achieved the highest pooled cumulative return (36.48%) and highest Sharpe ratio (1.13), outperforming all four

Table 7.1: Pooled out-of-sample test performance across 4 folds and 3 seeds (mean ± standard deviation over seeds).

Model	Cum. Return (%)	CAGR (%)	Sharpe Ratio	Sortino	Max DD (%)	Cost (% of V_0)
M1 (MLP)	20.62 ± 5.14	9.90 ± 2.31	0.71 ± 0.13	1.03 ± 0.20	-19.50 ± 1.04	9.68 ± 0.10
M2 (LSTM)	28.23 ± 2.03	13.37 ± 0.88	0.92 ± 0.07	1.34 ± 0.09	-18.52 ± 1.35	4.27 ± 1.44
M3 (+ DSR)	26.93 ± 3.27	12.68 ± 1.56	0.92 ± 0.11	1.34 ± 0.17	-19.95 ± 0.81	4.14 ± 1.15
M4 (+ Turn. Pen.)	20.97 ± 4.19	10.09 ± 1.85	0.92 ± 0.17	1.37 ± 0.26	$\mathbf{-16.04 \pm 1.72}$	$\mathbf{0.97 \pm 0.15}$
Equal-Weight	**36.48**	**17.08**	**1.13**	**1.66**	-18.91	**0.61**

reinforcement learning models. Among the active RL agents, M2 and M3 achieved the highest returns (28.23% and 26.93%), while M4 achieved the lowest execution friction (0.97% cost) and shallowest maximum drawdown (−16.04%).

Figure 7.1 illustrates cumulative portfolio profit across the continuous 2018–2025 timeline (with starting capital $V_0 =$ INR 1,000,000 reinvested sequentially), and Figure 7.2 portrays corresponding peak-to-trough drawdowns.

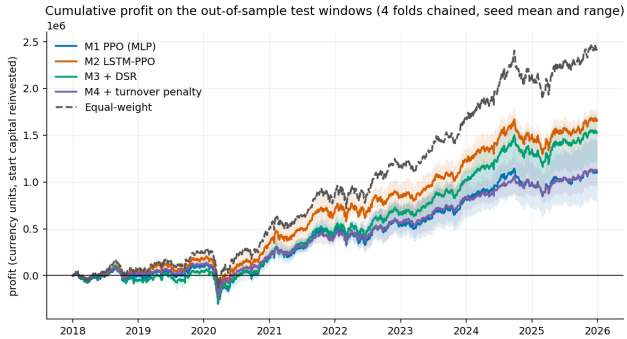


Figure 7.1: Cumulative out-of-sample profit across chained test windows (2018–2025, starting capital INR 1,000,000).

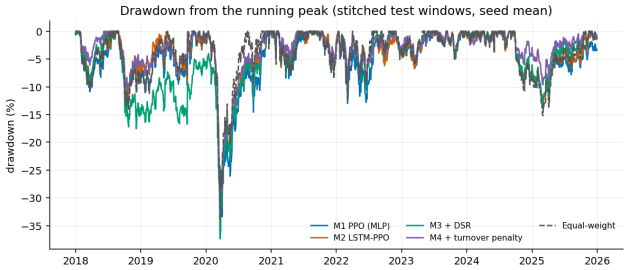


Figure 7.2: Out-of-sample portfolio drawdown from running peak across the chained timeline.

7.2 Analysis of Ablation Steps

To assess whether observed differences are consistent across environments, paired differences $(b - a)$ were computed across all 12 (fold, seed) evaluation pairs. Table 7.2 reports the paired mean deltas and 95% bootstrap confidence intervals (obtained by resampling whole folds with replacement across 10,000 bootstrap draws). These confidence intervals should be interpreted as finite-sample estimates reflecting the available 12 evaluation pairs rather than universal population parameters.

Table 7.2: Paired ablation differences $(b - a)$ with 95% bootstrap confidence intervals across 12 (fold, seed) pairs.

Ablation Transition	Metric	Mean Delta	95% CI	Folds Improved	Pairs Improved
M1 → M2 (Temporal Memory)	Sharpe Ratio	+0.201	[+0.156, +0.246]	4/4	9/12
	Cum. Return (%)	+7.61%	[+4.81%, +10.01%]	4/4	10/12
	Daily Turnover	−0.106	[−0.138, −0.070]	4/4	11/12
	Cost (% of V_0)	−5.41%	[−7.37%, −3.34%]	4/4	11/12
M2 → M3 (DSR Reward)	Sharpe Ratio	+0.003	[−0.295, +0.222]	3/4	7/12
	Cum. Return (%)	−1.30%	[−9.33%, +4.45%]	2/4	6/12
	Max Drawdown (%)	−1.43%	[−5.36%, +0.81%]	3/4	6/12
	Daily Turnover	−0.002	[−0.019, +0.014]	2/4	7/12
M3 → M4 (Turnover Regularisation)	Sharpe Ratio	+0.005	[−0.236, +0.280]	2/4	6/12
	Cum. Return (%)	−5.96%	[−13.29%, +3.37%]	1/4	3/12
	Max Drawdown (%)	**+3.90%**	[+1.45%, +7.00%]	**4/4**	**10/12**
	Daily Turnover	**−0.060**	[−0.082, −0.044]	**4/4**	**12/12**
	Cost (% of V_0)	**−3.17%**	[−4.01%, −2.43%]	**4/4**	**12/12**
M1 → M4 (All Combined)	Sharpe Ratio	+0.209	[+0.012, +0.386]	3/4	8/12
	Max Drawdown (%)	+3.45%	[+2.52%, +4.03%]	4/4	8/12
	Daily Turnover	−0.167	[−0.190, −0.144]	4/4	12/12
	Cost (% of V_0)	−8.71%	[−10.07%, −7.27%]	4/4	12/12

7.2.1 Step 1: Impact of Temporal Memory (M2 versus M1)

Transitioning from the MLP baseline (M1) to the LSTM recurrent encoder (M2) represents the clearest improvement in the ablation sequence. The pooled Sharpe ratio increases from 0.714 to 0.915 (+0.201, 95% CI [+0.156, +0.246]), improving across all four folds (4/4). Simultaneously, daily turnover decreases from 0.186 to 0.080, cutting cumulative transaction costs from 9.68% to 4.27% of capital.

The results suggest that the 30-day LSTM representation provides useful temporal information relative to the single-day MLP baseline under the tested configuration. Whereas the MLP reacts erratically to single-day price fluctuations, the recurrent state tracks multi-period trends, reducing counterproductive weight rebalancing.

7.2.2 Step 2: Impact of Differential Sharpe Ratio (M3 versus M2)

Replacing net return with the Differential Sharpe Ratio (M3) produces mixed, regime-dependent effects rather than a clear overall improvement. The pooled Sharpe ratio remains essentially flat (+0.003, 95% CI [−0.295, +0.222]), while pooled cumulative return decreases slightly from 28.23% to 26.93% and maximum drawdown widens from −18.52% to −19.95%. Although M3 achieves strong Sharpe ratios in Folds 3 and 4 (1.328 and 1.164), its recursive variance tracking dampens aggressive equity positioning during strong upward trends, slightly curtailing profit capture.

7.2.3 Step 3: Impact of Turnover Regularisation (M4 versus M3)

Augmenting the reward with a quadratic weight-change penalty ($\lambda_{\text{turn}} = 0.1$) demonstrates a distinct operational trade-off:

- **Substantial Friction Reduction:** Average daily turnover decreases from 0.079 to 0.019 (a 76% reduction, improved in 12/12 pairs), reducing transaction costs to 0.97% of starting capital, approaching the passive benchmark (0.61%).
- **Downside Protection:** Maximum drawdown improves by +3.90% (from −19.95% to −16.04%, 95% CI [+1.45%, +7.00%]), recording the shallowest drawdown among all models, including Equal-Weight (−18.91%).
- **Return Trade-off:** This conservatism reduces pooled cumulative return from 26.93% to 20.97% (improved in only 1/4 folds). Turnover regularisation successfully limits trading churn and drawdown, but penalises rapid reallocations needed to capture explosive market recoveries.

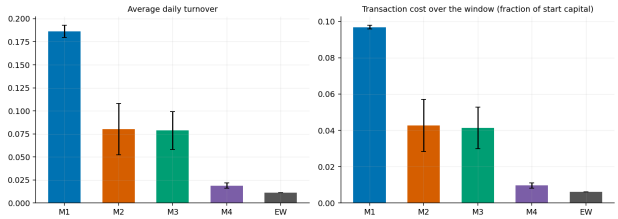


Figure 7.3: Average daily turnover and total transaction costs (% of initial capital V_0) across models.

7.3 Fold-by-Fold Performance Dynamics

Table 7.3 reports out-of-sample cumulative returns across each individual two-year test window, and Figure 7.4 illustrates cumulative profit trajectories over each fold.

Performance varies noticeably across market regimes:

Table 7.3: Out-of-sample cumulative return (%) per walk-forward fold (mean ± standard deviation over seeds).

Fold Window	Model M1	Model M2	Model M3	Model M4	Equal-Weight
Fold 1 (2018–2019)	10.0 ± 1.7	17.8 ± 7.7	4.4 ± 5.9	12.1 ± 5.3	**24.9**
Fold 2 (2020–2021)	34.0 ± 9.2	44.8 ± 6.9	44.1 ± 4.7	28.5 ± 14.8	**50.4**
Fold 3 (2022–2023)	23.4 ± 9.7	31.6 ± 4.7	34.5 ± 5.2	28.3 ± 5.3	**43.4**
Fold 4 (2024–2025)	15.0 ± 4.2	18.7 ± 1.3	24.7 ± 3.6	15.1 ± 4.0	**27.3**
Pooled Mean	20.6 ± 5.1	28.2 ± 2.0	26.9 ± 3.3	21.0 ± 4.2	**36.5**

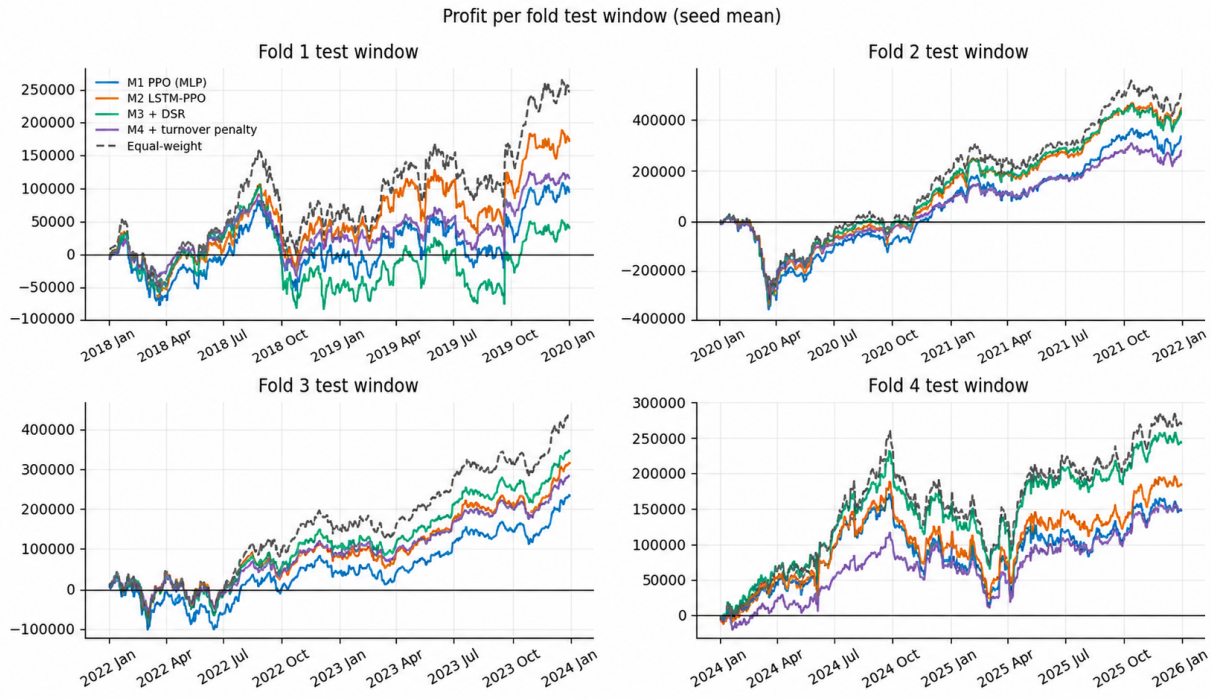


Figure 7.4: Cumulative out-of-sample portfolio profit across the four two-year test windows (mean across random seeds; initial capital $V_0 =$ INR 1,000,000; y-axis denotes profit in INR; January 2026 indicates the final axis tick for test data ending in December 2025).

- **Fold 1 (2018–2019):** Moderate growth regime. M2 leads the RL agents (17.8%), while M3 lags (4.4%). Equal-Weight leads at 24.9%.
- **Fold 2 (2020–2021):** Pandemic crash followed by strong liquidity-driven recovery. M2 (44.8%) and M3 (44.1%) perform similarly, whereas M4 lags (28.5%) because its weight-change penalty discouraged aggressive repositioning during the sharp recovery.
- **Fold 3 (2022–2023):** Inflationary tightening and sector rotation. M3 achieves its strongest performance (34.5%, Sharpe 1.328), outperforming M2 (31.6%) and M4 (28.3%).
- **Fold 4 (2024–2025):** Secular bull market. M3 delivers 24.7%, while M4 (15.1%) is constrained by conservative allocations.

The passive Equal-Weight benchmark produced higher cumulative returns than every RL model in all four test windows.

7.4 Equal-Weight Benchmark Outperformance and Gap Decomposition

To understand why the passive benchmark outperformed active RL, we examine two separate metrics: the **pooled average 2-year fold return** (36.48% in Table 7.1) and the **compounded 8-year return across the chained 2018–2025 timeline** (135.66% in Table 7.4).

Table 7.4 decomposes the chained cumulative return gap between each model and Equal-Weight into two exploratory components:

1. **Cash Drag:** The return forfeited by holding uninvested cash in an upward-trending market. A counterfactual "EW Same Cash" portfolio is constructed that mirrors each model's exact daily cash weight while equally weighting the remaining capital across the 10 equities.
2. **Stock Selection and Timing:** The difference between the model's return and the counterfactual "EW Same Cash" portfolio, reflecting active equity weights.

Table 7.4: Exploratory decomposition of cumulative out-of-sample return gap against Equal-Weight (% points over the chained 2018–2025 timeline).

Model	EW Return	Cash Drag	EW Same Cash	Stock Selection	Model Return
M1 (MLP)	135.66	−13.67	121.99	−2.11	119.88
M2 (LSTM)	135.66	−9.99	125.67	−2.14	123.53
M3 (+ DSR)	135.66	−15.14	120.52	−3.32	117.20
M4 (+ Turn. Pen.)	135.66	**−42.35**	93.31	−7.84	**85.47**

Table 7.5 reports the average portfolio allocations over the out-of-sample period.

Table 7.5: Average asset allocations across models over the out-of-sample evaluation period.

Model	RELI	TCS	HDFCB	ICICIB	ITC	HIND	LT	SUNP	MARU	BHAR	Cash
M1	0.088	0.120	0.085	0.087	0.084	0.077	0.091	0.087	0.087	0.094	**0.101**
M2	0.098	0.144	0.073	0.091	0.081	0.087	0.082	0.090	0.098	0.071	**0.087**
M3	0.089	0.119	0.073	0.077	0.072	0.079	0.073	0.094	0.107	0.086	**0.131**
M4	0.066	0.103	0.079	0.061	0.049	0.066	0.057	0.071	0.092	0.032	**0.324**
EW	0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100	**0.000**

The decomposition suggests that cash drag may account for a substantial portion of the return gap under the assumptions used:

- The 2018–2025 Indian equity market experienced strong overall growth. The passive Equal-Weight benchmark remained 100% invested across all ten large-caps at all times, capturing market beta with minimal turnover (0.61% total friction).

- Model M4 allocated an average of 32.4% to cash to avoid volatility and rebalancing penalties. Under the zero-return cash assumption, this cash buffer resulted in an estimated −42.35% cash drag relative to Equal-Weight, compared to −7.84% from equity selection.
- **Survivorship and Selection Bias:** The selected equities are established corporate leaders that prospered over 2011–2025. In a universe characterized by strong secular upward drift, passive diversification provides an effective baseline that active rebalancing strategies struggle to exceed after transaction frictions.

7.5 M4 Hyperparameter Tuning and Validation Dynamics

Figure 7.5 depicts the parameter sweep over $\lambda_{\text{turn}} \in \{0.01, 0.03, 0.1, 0.3\}$ on Fold 1 validation data across two seeds (seeds 0 and 1).

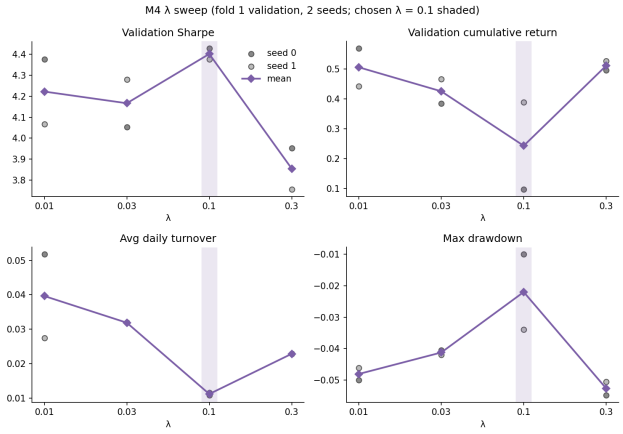


Figure 7.5: M4 turnover penalty parameter sweep on Fold 1 validation (chosen $\lambda = 0.1$ shaded).

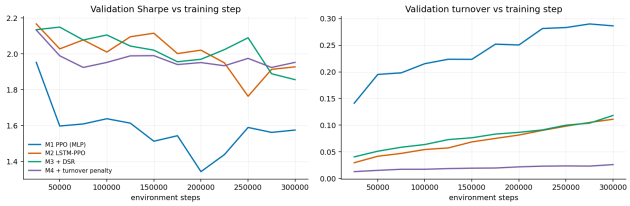


Figure 7.6: Validation Sharpe and turnover dynamics over 300,000 steps.

The preliminary sweep indicated that $\lambda = 0.1$ achieved a favorable trade-off on Fold 1 validation, reducing validation turnover while maintaining Sharpe ratio. At $\lambda = 0.01$, turnover remained elevated (0.040), whereas at $\lambda = 0.3$, reallocations were overly restricted. We disclose as an experimental limitation that $\lambda_{\text{turn}} = 0.1$ was selected based on preliminary validation tuning on Fold 1 alone across two seeds and applied frozen across all subsequent folds.

Finally, Figure 7.6 demonstrates that validation performance peaks between 50,000 and 100,000 steps before degrading, highlighting the necessity of checkpoint selection based on validation Sharpe ratio rather than relying on the final training step.

Chapter 8 — Limitations and Future Work

8.1 Methodological and Empirical Limitations

While this project establishes a controlled single-factor evaluation of reinforcement learning components for portfolio management, several important empirical and modeling limitations should be acknowledged:

1. **Survivorship and Selection Bias:** The 10 equities evaluated (e.g., Reliance, TCS, HDFC Bank, ICICI Bank) represent established large-cap market leaders that survived and prospered over the 2011–2025 period. In an asset universe characterized by strong long-term upward drift, a passive, fully-invested allocation captures broad market beta at minimal rebalancing friction (0.61%). Evaluating on a fixed set of historically successful equities introduces selection bias that inherently favors passive holding.

2. **Zero-Return Cash Assumption:** The portfolio environment models cash with an assumed zero nominal return. In reality, institutional cash holdings earn a risk-free rate or overnight yield (e.g., Indian MIBOR or Treasury bill yields). Assuming zero return on cash penalizes the nominal growth of risk-averse agents (especially Model M4, which held 32.4% in cash) during sustained bull markets.

3. **Simplified Execution and Market Friction Model:** Transaction costs are modeled as a constant proportional fee of $c_{\text{trans}} = 0.001$ (10 bps). While reasonable for liquid equities, this does not capture nonlinear market impact, order-book liquidity depletion, bid-ask spread widening during volatility, or intraday execution slippage.

4. **Limited Hyperparameter Tuning Scope:** The quadratic penalty parameter $\lambda_{\text{turn}} = 0.1$ was selected based on preliminary validation tuning on Fold 1 across two seeds, and kept frozen across all subsequent folds. A static penalty tuned in a relatively calm market regime (2017) may prove overly restrictive during periods of sudden market rebound (such as post-March 2020 in Fold 2).

5. **Sample Size of Evaluation Pairs:** The empirical evaluation relies on four walk-forward folds and three random seeds, yielding 12 paired observations. While bootstrap resampling provides informative confidence interval estimates, the modest number of market regimes limits broad statistical generalizations across arbitrary market conditions.

8.2 Directions for Future Research

The findings and trade-offs observed in this thesis point to several valuable directions for future investigation:

1. **Point-in-Time Dynamic Universes:** Constructing dynamic, point-in-time asset universes (such as historical Nifty 50 constituents reconstituted semi-annually) would mitigate survivorship bias and evaluate adaptability to corporate turnover.

2. **Interest-Bearing Cash Reserves:** Modeling an active risk-free interest rate yield on cash allocations would provide a fairer reflection of money-market returns and partially alleviate artificial cash drag.

3. **Benchmark-Relative Rewards:** Training policies against excess returns or tracking-error-penalized rewards relative to a market benchmark could encourage active exposure management while capturing index beta.

4. **Adaptive and Regime-Aware Regularisation:** Investigating dynamic turnover penalty schedules conditioned on real-time market volatility (e.g., India VIX) or validation-only tuning per fold.

5. **Alternative Temporal Encoders:** Exploring temporal Transformer backbones or state-space models (e.g., Mamba) to capture multi-scale cross-asset dependencies beyond single-stream recurrent cells.

Chapter 9 — Conclusion

9.1 Summary of Research Findings

This B.Tech. project developed, implemented, and empirically evaluated a continuous Actor–Critic Deep Reinforcement Learning framework for daily portfolio allocation across ten major equities on

the National Stock Exchange of India. By formulating the policy over a multivariate Dirichlet distribution, the agents generated continuous target allocations that strictly preserved the portfolio simplex manifold with an explicit cash reserve. Through a single-factor four-stage ablation study (M1–M4) evaluated across four rolling walk-forward folds and three random seeds (2018–2025), this study isolated the individual contributions of temporal memory, risk-sensitive rewards, and turnover regularisation under identical transaction frictions (0.1%).

9.2 Answers to Research Questions

Based on the empirical evidence gathered across the 12 evaluation pairs, we provide direct answers to the core research questions:

- **RQ1 (Temporal Architecture):** *Does an LSTM-based recurrent encoder improve out-of-sample risk-adjusted returns and policy stability compared to an MLP encoder?*
  Answer: Under our experimental setup, yes. Transitioning from the instantaneous MLP (M1) to the sequence-aware LSTM (M2) improved the pooled out-of-sample Sharpe ratio from 0.714 to 0.915 ($+0.201$, 95% CI $[+0.156, +0.246]$), with consistent gains across all four folds (4/4). Moreover, the LSTM reduced average daily turnover from 0.186 to 0.080 and cut total transaction costs from 9.68% to 4.27% of capital, suggesting that multi-day temporal representations help smooth short-term price noise and prevent erratic daily reallocations.
- **RQ2 (Risk-Adjusted Reward):** *Does transitioning from raw net return to the Differential Sharpe Ratio (DSR) lead to improved Sharpe ratios and reduced drawdowns?*
  Answer: The effect is mixed and regime-dependent. The DSR reward (M3) yielded an incremental Sharpe delta of $+0.003$ (95% CI $[-0.295, +0.222]$) relative to M2. While DSR showed strong performance in later regimes (Fold 3 Sharpe 1.328 and Fold 4 Sharpe 1.164), its online variance tracking slightly lowered cumulative return (26.93% vs. 28.23%) and did not improve pooled maximum drawdown (-19.95% vs. -18.52%).
- **RQ3 (Turnover Regularisation):** *Does augmenting the reward with an explicit quadratic penalty effectively suppress turnover and transaction costs, and what is the trade-off?*
  Answer: Yes, with a clear trade-off. Incorporating the quadratic weight-change penalty in Model M4 reduced average daily turnover from 0.079 to 0.019 ($[-0.082, -0.044]$, improved in 12/12 pairs), bringing total transaction costs down to 0.97% of starting capital. Crucially, M4 recorded the shallowest maximum drawdown among all evaluated models (-16.04%, an improvement of $+3.90\%$ over M3). However, this conservatism reduced cumulative return (20.97% vs. 26.93%), reflecting a trade-off between lower trading friction and capital appreciation during strong bull rallies.
- **RQ4 (Benchmark Competitiveness):** *Can a continuous actor-critic DRL agent outperform a passively rebalanced equal-weight benchmark in a realistic Indian equity universe?*
  Answer: No. The passive Equal-Weight benchmark achieved superior pooled cumulative return (36.48%) and Sharpe ratio (1.13). Our exploratory gap decomposition indicates that this deficit was primarily driven by *cash drag* (-42.35% in M4) arising from the agents' learned risk-averse cash allocations in a secular bull market, coupled with the zero-return cash modeling assumption. In an equity universe of established large-cap leaders characterized by strong upward drift, the market beta and low rebalancing cost (0.61%) of passive 1/N diversification represent a formidable benchmark for active reinforcement learning strategies.

9.3 Concluding Remarks

In conclusion, this B.Tech. project demonstrates that deep reinforcement learning can autonomously discover disciplined risk-management and cost-conscious trading policies in real financial markets.

While active outperformance against passive broad-market equities is bounded by cash drag and execution costs, the controlled ablation methodology established in this thesis provides a transparent, scientifically grounded baseline for future investigations in quantitative financial machine learning.

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